

Template PC / Golden Image Preparation

Internal IT Knowledge Base / Standard Operating Procedure (SOP)

Document Type	Internal IT Knowledge Base / User Guide / SOP
Audience	IT Administrators, IT Support, Help Desk, and Non-IT Staff
Applies To	Windows workstation deployment using WDS
Environment	Windows Server
Related Services	WDS01 - Windows Deployment Services (WDS)

1. Purpose

This document explains how to prepare a Template PC, also known as a Golden Image workstation, for organization-wide Windows deployment.

The Template PC is configured once with the organization's standard applications and workstation settings. It is then generalized using Sysprep, captured as a .wim image, and stored on the WDS server for deployment to other workstations.

The objective is to provide every newly deployed workstation with a consistent and standardized starting configuration.

2. Scope

This procedure covers:

- Creating the Template PC
- Installing Windows
- Installing standard applications
- Applying basic workstation configuration
- Preparing the system for imaging
- Running Sysprep
- Capturing the workstation through WDS
- Creating the organization's Golden Image
- Preparing the image for deployment

This procedure does not cover:

- Creating the Active Directory domain
- Creating users
- Creating Group Policies
- Building the DHCP server
- Building the DNS server

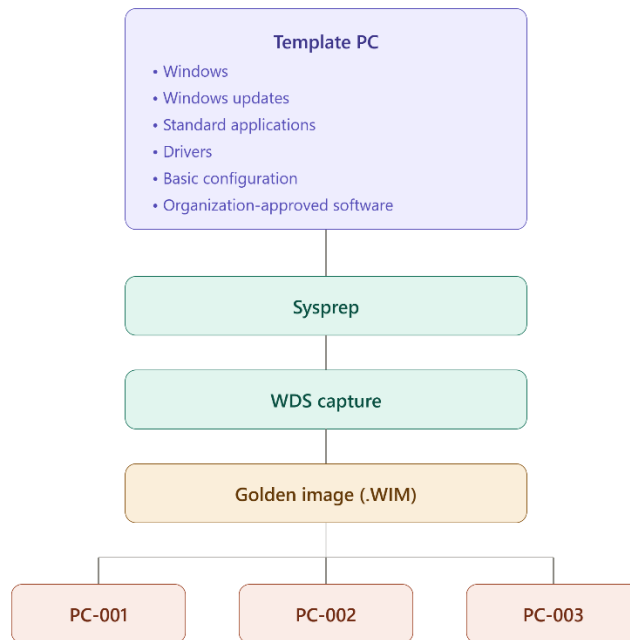
- Installing WDS
- Joining a deployed workstation to the domain

Those processes are documented separately.

3. What Is a Template PC?

A Template PC is a clean Windows workstation that IT configures to represent the organization's standard workstation.

For example:



Instead of manually installing and configuring every workstation, IT can deploy the same standardized image.

4. Responsibilities

IT Support / Deployment Technician

Responsible for:

- Preparing the Template PC
- Installing approved software
- Verifying the configuration
- Running pre-capture checks
- Running Sysprep
- Capturing the image
- Documenting the image version

System Administrator

Responsible for:

- Maintaining WDS
- Maintaining deployment infrastructure
- Managing Active Directory
- Managing Group Policy
- Approving standard workstation configurations
- Maintaining image versions

End Users

End users normally do not perform any part of this procedure.

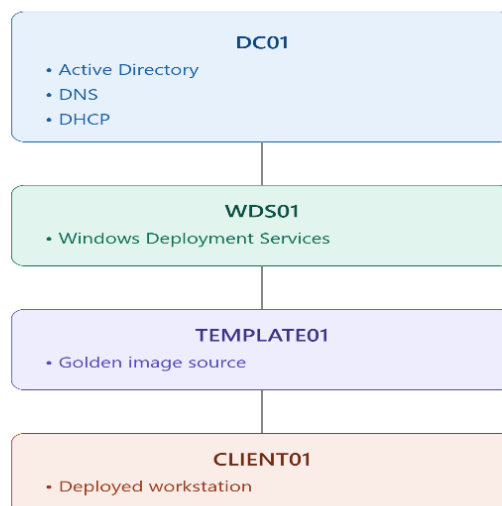
After deployment, users receive a workstation that has already been prepared by IT.

5. Prerequisites

Before creating the Template PC, verify that the following infrastructure is operational.

Component	Required
Active Directory	Yes
DNS	Yes
DHCP	Required for PXE deployment
WDS	Yes
Windows installation media	Yes
Network connectivity	Yes
Windows Pro/Enterprise edition	Recommended
WDS storage	Yes

For your lab environment, the typical architecture is:



6. Template PC Standards

The Template PC should use hardware or a VM configuration representative of the organization's target workstations.

Recommended lab configuration:

Component	Recommended
Computer Name	TEMPLATE01
OS	Windows 10 Pro
RAM	4 GB or more
CPU	2 cores or more
Storage	40–60 GB or more
Network	Internal/Lab network
Domain	WORKGROUP
IP	DHCP

Important: The Template PC should not be joined to the production/domain environment before Sysprep.

7. Procedure

Step 1 — Create the Template PC

Create a new physical workstation or virtual machine.

For a lab environment:

- Name: TEMPLATE01
- OS: Windows 10 Pro
- RAM: 4 GB
- CPU: 2 cores
- Disk: 40–60 GB
- Network: labnet

Ensure the workstation can communicate with the WDS server.

8. Step 2 — Install Windows

Install the organization's approved Windows version.

During installation:

- Boot from Windows installation media.
- Select the appropriate Windows edition.
- Partition the disk as required.
- Complete Windows installation.
- Log in using the local administrator account.

Important: Do not join the computer to Active Directory at this stage.

Verify:

Settings
→ System
→ About

The machine should still belong to:

WORKGROUP

9. Step 3 — Install Windows Updates

Before capturing the image, fully update Windows.

Go to:

Settings
→ Update & Security
→ Windows Update

Install available updates.

Restart the computer when required.

Continue checking for updates until the workstation is fully updated according to the organization's approved image baseline.

10. Step 4 — Install Standard Applications

Install only applications approved for the standard workstation image.

Examples:

- Microsoft Edge
- Google Chrome
- 7-Zip
- PDF reader
- Microsoft Office, if licensed and appropriate
- Organization-approved utilities
- Required runtime components
- Approved endpoint/security software

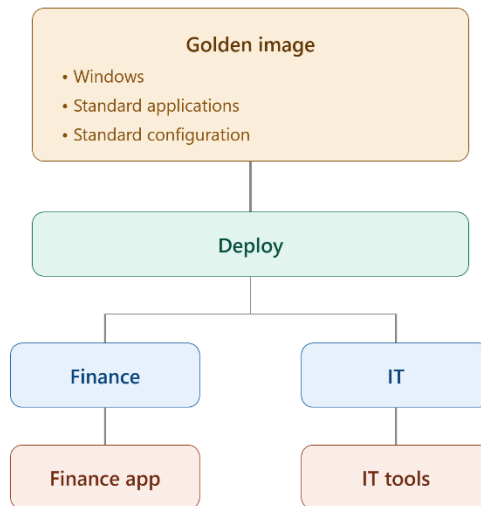
Avoid installing:

- Personal applications
- Temporary troubleshooting tools
- Unapproved software
- Applications intended only for specific employees
- User-specific configuration

Example

Do not put an employee-specific application into the Golden Image if only one department uses it.

Instead:



Department-specific software can be installed later through IT processes, software deployment, or Group Policy where appropriate.

11. Step 5 — Install Drivers

Install the appropriate drivers for the target hardware.

At minimum, verify:

- Network adapter
- Storage controller
- Chipset
- Graphics
- Audio
- Other organization-required hardware

For a WDS environment supporting multiple hardware models, IT should consider whether drivers should be maintained separately in WDS rather than permanently embedded in the image.

12. Step 6 — Apply Basic Workstation Configuration

Configure settings that should be common across all workstations.

Examples:

- Power configuration
- Standard Windows settings
- Approved applications
- Basic desktop configuration
- Organization-approved utilities
- Basic troubleshooting tools

However, avoid putting centralized domain policies directly into the image.

13. Image vs. Group Policy

A common mistake is trying to configure everything inside the Template PC.

Use the following principle:

Template Image

Contains:

- Windows
- Applications
- Drivers
- Basic workstation configuration

Active Directory / Group Policy

Controls:

- Password policies
- Security policies
- USB restrictions
- Control Panel restrictions
- Windows Update policies
- Desktop policies
- Network drive mappings
- Printer deployment
- User restrictions
- Computer restrictions

This allows IT to change policies centrally without rebuilding the Golden Image every time.

14. Step 7 — Verify the Template PC

Before Sysprep, perform a complete verification.

Operating System

Verify:

- Correct Windows edition
- Windows is activated according to organizational licensing
- Windows Updates are complete
- No obvious Windows errors

Applications

Open each standard application and verify that it works.

Network

Run:

```
ipconfig /all
```

Confirm that the workstation has network connectivity.

Domain Status

Verify that the computer is not domain joined.

Run:

```
systeminfo | findstr /B /C:"Domain"
```

Expected:

```
Domain: WORKGROUP
```

15. Step 8 — Clean the Template PC

Before capturing the image:

Remove unnecessary:

- Temporary files
- Installation files
- Downloaded installers
- Temporary user data
- Test files
- Test accounts
- Troubleshooting artifacts

Empty the Recycle Bin.

Important: Do not delete system directories or Windows default profiles manually.

16. Step 9 — Pre-Sysprep Checklist

Before running Sysprep, confirm:

- ☐ Windows installation is complete
- ☐ Windows Updates are complete
- ☐ Standard applications are installed
- ☐ Drivers are installed
- ☐ Basic configuration is complete
- ☐ Temporary files are removed
- ☐ Test accounts are removed
- ☐ Computer is not domain joined
- ☐ Computer is connected to the correct network
- ☐ WDS server is reachable
- ☐ Image storage is available
- ☐ Important data has been removed
- ☐ Template configuration has been documented

17. Step 10 — Run Sysprep

Sysprep prepares Windows for deployment to other computers.

Open:

```
C:\Windows\System32\Sysprep
```

Run:

```
sysprep.exe
```

Select:

System Cleanup Action:
Enter System Out-of-Box Experience (OOBE)

Generalize:
Enabled

Shutdown Options:
Shutdown

The final configuration should be:

```
System Cleanup Action
Enter System Out-of-Box Experience

☒ Generalize

Shutdown Options
Shutdown
```

Click OK.

18. Critical Sysprep Rule

After Sysprep finishes:

Important: Do not boot the Template PC back into Windows before capturing the image.

The system should remain powered off.

The next step is to boot the computer through the WDS/PXE environment and perform the image capture.

19. Step 11 — PXE Boot

Start the Template PC.

Boot from the network/PXE interface.

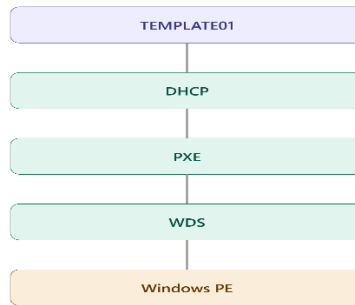
Depending on the system, this may require:

```
F12 or Esc
```

or another boot-menu key.

The workstation should obtain network information from DHCP and contact the WDS server.

Expected flow:



20. Step 12 — Capture the Image

From the WDS capture environment:

- Select the appropriate Windows installation.
- Select the Windows partition.
- Enter an image name.
- Enter a description.
- Select the destination for the .WIM file.
- Start the capture.

Recommended image naming:

```
Windows10-Pro-Standard-v1.0.wim
```

Example description:

```
Windows 10 Pro standard workstation image.
Updated August 2026.
```

21. Step 13 — Store the Golden Image

Store the captured image on the designated WDS image storage.

Example:

```
WDS01
├── Images
│   └── Windows10
│       └── Windows10-Pro-Standard-v1.0.wim
```

Maintain image versions.

Example:

```
v1.0
v1.1
v1.2
v2.0
```

Important: Do not overwrite an existing production image without following the organization's change-management process.

22. Image Version Management

Each image should have documentation.

Field	Example
Image Name	Windows10-Pro-Standard
Version	1.0
Created	August 2026
OS	Windows 10 Pro
Applications	Standard workstation applications
Windows Updates	Current at creation
Created By	IT Deployment
Status	Approved
WDS Server	WDS01

When an image is modified, increment the version.

Example:

- 1.0 → Initial Golden Image
- 1.1 → Updated Chrome
- 1.2 → Added approved PDF reader
- 2.0 → Major Windows/application baseline update

23. Step 14 — Add Image to WDS

On the WDS server:

```
Windows Deployment Services
    ↓
Servers
    ↓
WDS01
    ↓
Install Images
```

Create/select the appropriate image group.

Import the captured .WIM file.

Example:

```
Install Images
├── Windows 10
│   └── Windows 10 Pro - Standard Workstation
```

24. Step 15 — Test the Golden Image

Important: Never immediately deploy a newly created image to production workstations without testing it.

Deploy it first to a test client:

CLIENT-TEST01

Verify:

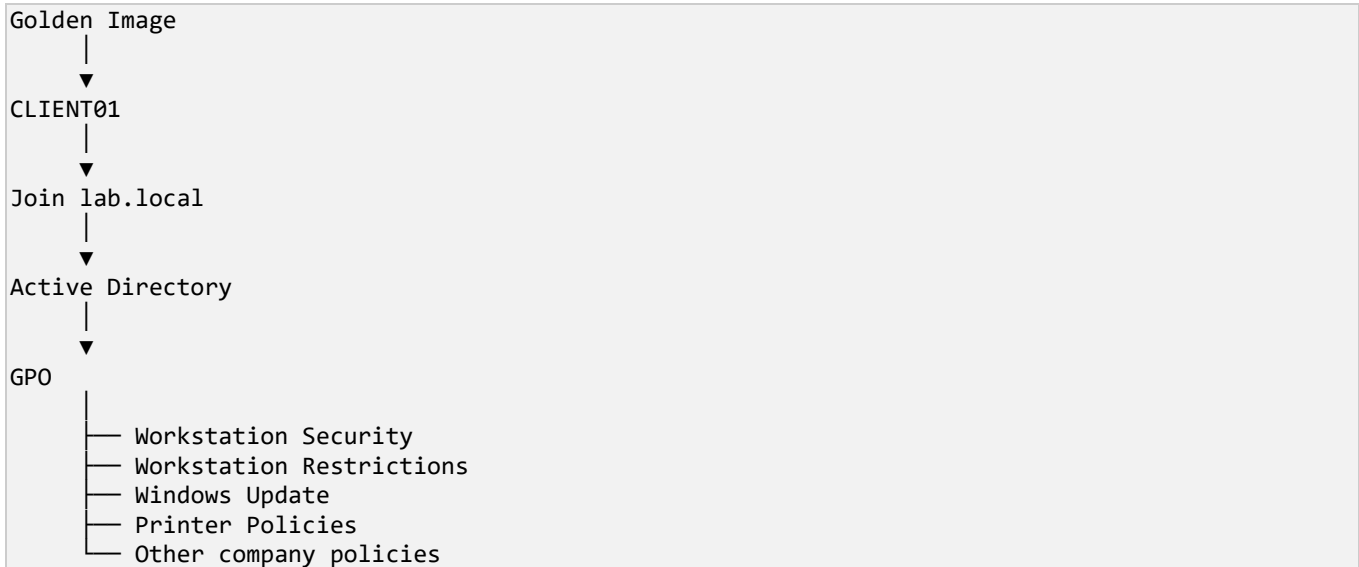
- Windows boots correctly
- Applications work
- Network connectivity works
- Windows activation/licensing behaves as expected
- The computer can join the domain
- Group Policy applies
- DNS works
- DHCP works
- Required printers work
- Required network drives work
- Standard users can log in
- Security policies work

25. Domain Join After Deployment

The Template PC itself should normally remain:

WORKGROUP

After deployment:



This is the preferred architecture for your lab.

26. What End Users Need to Know

Non-IT users generally do not need to perform any Template PC configuration.

When receiving a new workstation, users should simply:

- Connect the computer to the organization's network.
- Turn on the workstation.
- Sign in using their organization account.
- Allow required policies and applications to finish configuring.
- Contact IT if something is missing or not working.

Users should not:

- Run Sysprep
- Modify the Golden Image
- Change deployment settings
- Modify WDS
- Change DNS settings
- Remove security software
- Join or remove computers from the domain without authorization

27. Troubleshooting

WDS does not appear during PXE boot

Check:

```

DHCP
↓
Network connectivity
↓
PXE
↓
WDS

```

Verify:

- Client receives an IP address
- Client is connected to the correct network
- WDS service is running
- DHCP is working
- PXE boot is enabled
- VM network adapter is connected to the correct internal network
- WDS server is reachable

Sysprep fails

Check:

```
C:\Windows\System32\Sysprep\Panther
```

Look for:

```
setupact.log
setuperr.log
```

Common causes include:

- Unsupported application state
- Windows Store/UWP application problems
- Previous Sysprep attempts
- Corrupted Windows installation
- Incorrect system configuration

Important: Do not repeatedly run Sysprep without identifying the underlying problem.

Template boots into Windows after Sysprep

If the machine was accidentally booted after Sysprep, stop and verify the image state before continuing with capture.

Do not immediately assume that the resulting image is suitable for deployment.

Client receives DHCP but WDS does not start

Verify:

Client

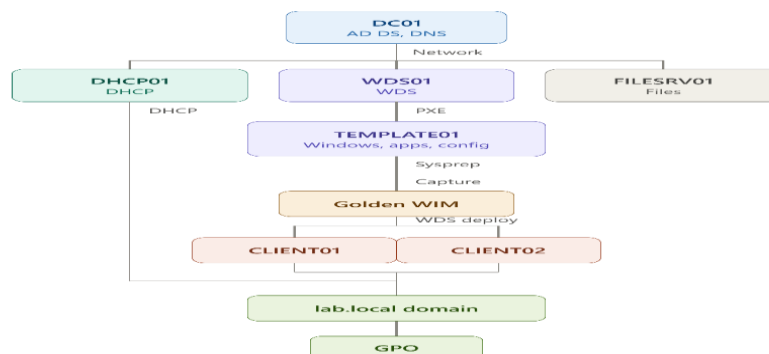
- DHCP → Working
- PXE/WDS → Not working

Focus troubleshooting on:

- PXE configuration
- WDS
- Network boot configuration
- Firewall
- DHCP/WDS integration
- BIOS/UEFI boot mode

28. Recommended Deployment Architecture

For the organization's lab:



29. Golden Image Best Practices

Keep the image minimal

Only include software that should exist on nearly every workstation.

Use Group Policy for centralized configuration

Don't rebuild the image whenever a security policy changes.

Keep image versions

Never lose track of which image is currently approved.

Test before production deployment

Always deploy a new image to a test machine first.

Document changes

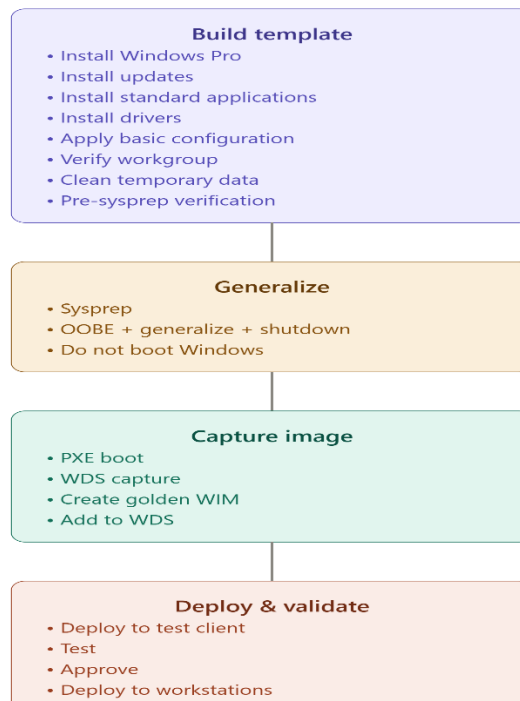
Record:

- What changed?
- Why did it change?
- Who changed it?
- When was it changed?
- What image version contains the change?

Keep a known-good image

Maintain the previous approved image until the new image has been validated.

30. Quick Reference



31. Final Acceptance Checklist

Template PC

- ☐ Correct Windows edition installed
- ☐ Windows Updates completed
- ☐ Required drivers installed
- ☐ Standard applications installed
- ☐ Basic workstation configuration completed
- ☐ Temporary files removed
- ☐ Test accounts removed
- ☐ Computer remains in WORKGROUP
- ☐ No user-specific information remains
- ☐ Network connectivity verified
- ☐ WDS connectivity verified

Sysprep

- ☐ Sysprep completed successfully
- ☐ Generalize selected
- ☐ OOBE selected
- ☐ Shutdown selected
- ☐ Template PC powered off
- ☐ Template PC was not booted into Windows after Sysprep

WDS

- ☐ Template successfully PXE booted
- ☐ Windows PE loaded
- ☐ Correct Windows partition selected
- ☐ WIM capture completed
- ☐ Image stored on WDS
- ☐ Image imported into WDS
- ☐ Image version documented

Testing

- ☐ Test client deployed
- ☐ Windows boots successfully
- ☐ Applications work
- ☐ Network works
- ☐ Domain join works
- ☐ DNS works
- ☐ GPO applies
- ☐ User login works